B.Tech Petroleum Engineering

MS: Mid Semester | IA: Internal Assessment | ES: End Semester | LW: Laboratory Work | LE/Viva: Laboratory Examination/Viva | CE: Continuous Evaluation

Semester I													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24MA101T	Mathematics-I	3	1	0	4	4	**25**	**25**	**50**	--	--	100
24CP101T	Computer Programming-I	1	0	0	1	1	25	25	50	--	--	100
24CP101P	Computer Programming-I Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PH103T	Modern Physics	3	0	0	3	3	25	25	50	--	--	100
24PH103P	Modern Physics Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CV101T	Environmental Science	2	0	0	2	2	25	25	50	--	--	100
24ME101P	Workshop Practice	0	0	2	2	1	--	--	--	50	50	100
24ME102P	Engineering Graphics	0	0	4	4	2	--	--	--	50	50	100
24PE101T	Geology for Petroleum Engineers	3	0	0	3	3	25	25	50	--	--	100
24PE101P	Geology for Petroleum Engineers Practical	0	0	2	2	1	--	--	--	50	50	100
24HS101T	English Communication	2	0	2	2	2	25	25	50	--	--	100
24HS102T	Universal Human Values	1	0	0	1	1	25	25	50	--	--	100
	Total	**15**	**1**	**12**	**28**	**22**						

Semester II													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24MA103T	Mathematics-II	3	1	0	4	4	25	25	50	--	--	100
24CP102T	Computer Programming-II	1	0	0	1	1	25	25	50	--	--	100
24CP102P	Computer Programming-II Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH102T	Chemistry	3	0	0	3	3	25	25	50			100
24CH102P	Chemistry Practical	0	0	2	2	1	--	--	--	50	50	100
24PE102T	Sedimentary and Petroleum Geology	3	0	0	3	3	25	25	50	--	--	100
24PE102P	Sedimentary and Petroleum Geology Practical	0	0	2	2	1	--	--	--	50	50	100
24HS103T	Indian Knowledge System	2	0	0	2	2	25	25	50	--	--	100
24HS104T	Organizational Behaviour	1	0	0	1	1	25	25	50	--	--	100
24HS105T	Professional Communication	2	0	0	2	2	25	25	50	--	--	100
24YOG101	Yoga, Health & Hygiene OR NCC/NSS											
24NSS101	National Service Scheme (NSS)	0	0	2	2	1	--	--	--	50	50	100
24NCC101	National Cadet Corps (NCC)											
	Total	**15**	**1**	**8**	**24**	**20**						

24MA101T					Mathematics - I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- To be able to evaluate problems related to differential and integral calculus of complex functions.
- To be able to obtain area, volume using integral calculus.
- To be able to formulate and solve various engineering problems using the calculus.
- To study the properties of Matrix algebra and apply them to solve system of algebraic equations.

Unit I: Differential Calculus and its Applications **Hours: 12**

Partial derivative and its application, - Euler's theorem - Total derivatives - Jacobians – Maxima and Minima of two variables using Lagrange's multipliers. Convergence of power series.

Unit II: Integral Calculus and its Applications **Hours: 14**

Definition Evaluation of double integral (Cartesian – Polar form) –Change of orders - Change of variables – Evaluation of triple integral, change of variables (Cartesian to spherical – and cylindrical) – Applications, area – volume –center of mass- center of gravity by double and triple integral.

Unit III: Matrix Algebra and its Applications **Hours: 14**

Solution of system of algebraic equation - Rank of a matrix, consistency of system of equation - Characteristic equation of a square matrix- Eigen values and Eigenvectors of a real matrix - Properties of eigen values and eigen vectors - Cayley-Hamilton theorem (without proof) - finding inverse of a matrix- Diagonalization of a matrix using orthogonal transformation.

Unit IV: Vector Calculus **Hours: 12**

Gradient, divergence and curl – Directional derivative – Irrotational and Solenoidal vector fields – Vector Integration – Simple problems on line, surface and volume integrals – Green's theorem in a plane, Gauss divergence theorem and Stokes' theorem (without proofs) – Simple application involving cubes and rectangular parallelepipeds.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the use of convergence of infinite series in engineering aspects.
CO2 - Understand the concept of Directional derivative, Irrotational and Solenoidal vector fields.
CO3 - Apply appropriate tool/method to extract the solutions of engineering problems.
CO4 - Analyze the obtained solution in context with theory.
CO5 - Appraise mathematical problems from real to complex domain.
CO6 - Evaluate problems on Green's, Stokes' and Divergence theorems.

TEXT / REFERENCE BOOKS

1. Grewal, B. S. (2014). *Higher engineering mathematics* (43rd ed.). Khanna Publishers.
2. Jain, R. K., & Iyengar, S. R. K. (2007). *Advanced engineering mathematics* (3rd ed.). Alpha Science International.
3. Kreyszig, E. (2015). *Advanced engineering mathematics* (10th ed.). John Wiley & Sons.
4. Strang, G. (2005). *Linear algebra and its applications* (4th ed.). Cengage Learning.
5. Hoffman, K., & Kunze, R. A. (2002). *Linear algebra*. Prentice Hall of India.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24CP101T					Computer Programming – I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the usage of operators and data types.
- To apply different types of Conditional and looping statement.
- To create different types of data collections.
- To implement user defined function.
- To perform different operations upon files

Unit I: Basics of Programming **Hours: 3**

Introduction to Computer Programming, Features of C language, Structure of C program, program execution flow, C Tokens, variables, Data types, Operators, Decision control statements-if, switch, go to statement. Loop control structures- while, do-while, for loop, Break statement, Continue statement

Unit II: Derived Data Types **Hours: 4**

Array: one dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, String-Basic Concepts, Inbuilt String manipulation Functions, Pointer, Pointer arithmetic, Pointer to pointer, Array of Pointers

Unit III: Functions and Structures **Hours: 3**

Introduction to user defined functions, Types of Functions, Call by value-call by reference, recursion, pointers to functions, Structures, Array of Structure, Union

Unit IV: Files Handling **Hours: 3**

File handling in C, Different types of files, Operations on Files such as File creation, File deletion, File access modes such as read, write, append, File concatenation, File handling using seek function.

Max <13Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand basics of programming.
CO2 - Identify different programming constructs for a problem.
CO3 - Apply appropriate derived data type for data storage.
CO4 - Prepare a user defined data type based on data attributes.
CO5 - Construct user defined functions for problem solving.
CO6 - Analyse different data structure based on application requirement.

TEXT / REFERENCE BOOKS

1. Kernighan, B. W., & Ritchie, D. M. (1988). *The C programming language* (2nd ed.). Prentice Hall.
2. King, K. N. (2008). *C programming: A modern approach* (2nd ed.). W. W. Norton & Company.
3. Griffiths, D., & Griffiths, D. (2012). *Head first C: A brain-friendly guide*. O'Reilly Media.
4. Balagurusamy, E. (2019). *Programming in ANSI C* (8th ed.). McGraw-Hill Education.
5. Kanetkar, Y. P. (2016). *Let us C* (15th ed.). BPB Publications.
6. Kanetkar, Y. P. (2003). *Pointers in C*. BPB Publications.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: 2 Questions from each unit with internal choice, each carrying 20 marks **80 Marks**

24CP101P					Computer Programming-I Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To implement basic programming concepts.
- To create different types of data collections.
- To implement user defined function.
- To perform different file handling operations.

LIST OF EXPERIMENTS

1. **Introduction to Computer Programming**: Understanding compilation process through a simple C program, program execution flow, C Tokens, variables and keywords and identifiers, types of C constants and variables.
2. **Simple and formatted Input Output Operations:** Data types, Operators, Input /output statements in C, Formatted I/O, format specifiers, escaper sequences
3. **Decision making and branching:** if, if-else, if-else ladder, switch, go to statement, conditional operator statement
4. **Looping control structures:** while, do-while, for loop, Break statement, Continue statement
5. **Derived Data Type: Array and Strings:** One dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, Basic Concepts, Inbuilt String manipulation with and without using inbuilt functions.
6. **Derived Data Type: Structure and Union:** structure, arrays and structures, structures and functions, pointer to structure, typedef, unions
7. **Functions:** Introduction to user defined functions, Types of Functions, Call by value-call by reference, header file creation, recursion, pointers to functions, arrays and functions
8. **Pointers:** Pointer's basics, use of &, * operator in context to pointers, Pointer arithmetic, Array and String processing using pointer, pointer to pointer, Array of Pointers
9. **File Handling in C:** File handling in C, Different types of files, Operations on Files, File handling functions.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the use of appropriate naming conventions and programming style including appropriate comment density.
CO2 - Implement a basic C program using appropriate control structure.
CO3 - Apply appropriate derived data types based on data attributes.
CO4 - Develop solutions with pointers and utilize them to access strings and structures.
CO5 - Design user defined functions for problem solving and reuse them acccross different programs.
CO6 - Evaluate problems on Green's, Stokes' and Divergence theorems.

TEXT / REFERENCE BOOKS

1. Kernighan, B. W., & Ritchie, D. M. (1988). The C programming language (2nd ed.). Prentice Hall.
2. King, K. N. (2008). C programming: A modern approach (2nd ed.). W. W. Norton & Company.
3. Griffiths, D., & Griffiths, D. (2012). Head first C: A brain-friendly guide. O'Reilly Media.
4. Balagurusamy, E. (2019). Programming in ANSI C (8th ed.). McGraw-Hill Education.
5. Kanetkar, Y. P. (2016). Let us C (15th ed.). BPB Publications.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24PH103T					Modern Physics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- Understanding concepts of modern physics.
- Explaining the physics of EM waves and its propagation.
- Applying the Acquired basic knowledge of solid state physics.
- Analyze the Concepts of nuclear radiation physics
- Estimate and Correlate the concepts learned so far for various engineering applications
- Apply the fundamentals designs of modern physics to solve complex physical problems.

Unit I: Modern Physics **Hours: 09**

Review of quantum concepts: particle nature of light, photoelectric effect, Compton effect, matter waves, wave packets, phase and group velocity, Interference, Diffraction and Polarization, Engineering Physics related applications. Davisson Germer experiment, Heisenberg uncertainty principle.

Unit II: Electromagnetic Waves **Hours: 11**

Physical and mathematical concepts of gradient, divergence and curl, Gauss theorem, applications in gravitation and electrostatics. Stokes' theorem and engineering Physics related applications. Equation of continuity, Biot Savart law – Ampere's law – magnetization and magnetic intensity, Faraday's law of Maxwell's equations, wave equation for electromagnetic radiation, applications of optical fibers in communication.

Unit III: Solid State Physics **Hours: 11**

Crystalline and Amorphous Solids, Ionic Crystals, Covalent Crystals, Van der Waals Bond, Band Theory of Solids, Semiconductor Devices, Electrical conductivity, Resistivity, Magnetism, Superconductivity, Introduction to BCS Theory. Concepts of LASER, Interaction of radiation of matter-quantum mechanical view, characteristics and Types of laser, Engineering Physics related application of lasers.

Unit IV: Vector Calculus **Hours: 08**

Mass defect, binding energy, Radioactivity, Types of Radiation, Interaction of Radiation with matter, Radiation detector, nuclear reactions, elements of nuclear reactors, fission and fusion, Engineering Physics related problems**.**

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - To relate to shape the engineering perspective in the student mind.
CO2 - Aims to provide an understanding to analyse the physical phenomena of various physics concepts.
CO3 - To develop an analytical perspective in the student.
CO4 - To enable understanding in the students the importance of application of already studied topics
CO5 - To explain and relate the importance of interdisciplinary problems.
CO6 - To strengthen problem solving attitude in physics using mathematical tools.

TEXT / REFERENCE BOOKS

1. Resnick, R., Halliday, D., & Krane, K. S. (1992). *Physics* (Vols. 1–2, 4th ed.). John Wiley & Sons.
2. Ghatak, A. K. (2017). *Optics* (6th ed.). McGraw-Hill Education.
3. Purcell, E. M., & Morin, D. J. (2013). *Electricity and magnetism* (3rd ed.). Cambridge University Press.
4. Kittel, C., Knight, W. D., & Ruderman, M. A. (1973). *Mechanics* (2nd ed.). McGraw-Hill.
5. Griffiths, D. J. (2017). *Introduction to electrodynamics* (4th ed.). Cambridge University Press.
6. Avadhanulu, M. N., & Kshirsagar, P. G. (2019). *A textbook of engineering physics*. S. Chand & Company Ltd.
7. Feynman, R. P., Leighton, R. B., & Sands, M. (2010). *The Feynman lectures on physics* (Vol. 1). Narosa Publishing House.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PH103P					Modern Physics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- Understand the characteristics of waves, semiconductor, heat pump, LASER.
- Enhance knowledge on application of physics in engineering
- To develop intellectual communication skills and discuss the basic understanding of various experimental principles involved
- Demonstrate practical knowledge by applying experimental methods to correlate with the theory.
- Apply the analytical techniques and graphical analysis to the experimental data

LIST OF EXPERIMENTS

1. Determining Plank's constant and inverse square law
2. Study of Photoconductivity
3. Study of Bio-Savart's Law
4. Determining e/m by Thomson's method
5. Study of Hall Effect.
6. Experiments on single and double slit diffraction and interference with He-Ne Laser
7. Study of I-V characteristics of p-n diode.
8. Determination of thermal conductivity of different solids
9. To measure resistivity of semiconductor by Four Probe method and determination of band gap.
10. Study of Interference using Newton's Ring experiment.
11. To study G.M. tube characteristics and to calculate the dead time.
12. Energy calibration of CsI:Tl radiation detector and energy analysis of an unknown gamma source
13. To determine the numerical aperture of a given fibre optics cable using the far field measurements.
14. Experiments with heat pump
15. Study of Polarization of light using LASER

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Analysis the engineering problems and design the components for the solution.
CO2 - Developing skills to utilize the different tools for engineering problems.
CO3 - Analyse the results and correlate with theory and its application in industries.
CO4 - Design the set-up and utilize for component analysis.
CO5 - Identifying the problem and creating the solutions for research and development.
CO6 - Analyse the scientific data and learn to be efficient as individual and a team member.

TEXT / REFERENCE BOOKS

1. Runyan, W. R. (1975). *Semiconductor measurements and instrumentation*. McGraw-Hill.
2. Sayer, M., & Mansingh, A. (2000). *Measurement, instrumentation, and experiment design in physics and engineering*. Prentice Hall India.
3. Melissinos, A. C., & Napolitano, J. (2003). *Experiments in modern physics* (2nd ed.). Academic Press.
4. Nakra, B. C., & Chaudhry, K. K. (2009). *Instrumentation, measurement, and analysis* (3rd ed.). Tata McGraw-Hill.
5. ORTEC. (2006). *Experiments in nuclear science: Laboratory manual*. ORTEC.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Evaluation based on the class performance and Laboratory book **50 Marks**
End semester examination and Viva-voce **50 Marks**

24CV101T					Environment Science					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- To develop a comprehensive perspective of environment and sustainable development
- To understand the causes and effects of various types of pollution
- To develop an understanding of the various strategies for controlling the pollution
- To introduce the emerging environmental domains

Unit I: Introduction to Environment **Hours: 05**

Sustainable Development; Sustainable Development Goals; Environmental Studies – Its importance and Multidisciplinary nature, Introduction to Environmental Parameters and their standards (air, water, soil, noise, etc.); Ecosystem and its types, Ideal ecosystem, Biodiversity : Its importance and conservation.

Unit II: Multi-Scale Environmental Pollution (Global, Regional and Local) **Hours: 06**

Pollution, Causes and Effects of different types of pollution : Air Pollution, Water Pollution, Soil Pollution, Solid Waste (organic and Inorganic) Pollution, Hazardous Waste Pollution, Marine Pollution, Noise Pollution, Thermal Pollution, Radioactive Pollution; Introduction to man-made disasters like floods, heat waves, landslides, etc., Introduction to the various instruments for measuring air pollution, water pollution, noise, etc.

Unit III: Environmental Pollution Control Strategies **Hours: 08**

Multi-approaches for reducing various types of pollution: Introduction to Water and Wastewater treatment technologies, Air and Noise pollution control techniques, Introduction to different environmental management concepts like Swachh Bharat Mission, Mission LiFE (Lifestyle for Environment), etc. Indian Culture and Traditional Wisdom for managing environment

Unit IV: Emerging Environmental Managements Domains **Hours: 07**

Concept of Zero Liquid Discharge (ZLD) and the reuse of the treated wastewater, Green Credit Rules - 2023, Clean Development Mechanisms (CDM) and Carbon Credits, Green Buildings, Carbon Footprint and Water Footprint, Green Business, International Environmental Laws, Environmental Auditing

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Demonstrate comprehension of sustainable development and environmental aspects.
CO2 - Recognize the interdisciplinary characteristics inherent in Environmental studies.
CO3 - Evaluate the impact of various pollutants on the environment.
CO4 - Assess the efficacy of different technologies for environmental pollution control.
CO5 - Analyze different environmental management policies and their implications.
CO6 - Synthesize knowledge about emerging environmental management paradigms.

TEXT / REFERENCE BOOKS

1. Bharucha, E. (2005). *Textbook of environmental studies for undergraduate courses*. Universities Press.
2. Botkin, D. B., & Keller, E. A. (2014). *Environmental science: Earth as a living planet* (9th ed.). Wiley.
3. Miller, G. T., Jr. (2006). *Environmental science: Working with the Earth* (11th ed.). Cengage Learning.
4. Rajagopalan, R. (2016). *Environmental studies: From crisis to cure* (3rd ed.). Oxford University Press.
5. Masters, G. M., & Ela, W. P. (2012). *Introduction to environmental engineering and science* (3rd ed.). PHI Learning.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24ME101P					Workshop Practices					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To understand safety in various manufacturing processes
- Learn how to use various measuring tools for engineering applications
- Hands on training and preparation of job using wood, metal and sheet as per drawing
- Understand various manufacturing processes like machining, welding, soldering and 3D printing for prototypes.

LIST OF EXPERIMENTS

1. Introduction to Workshop safety, layout and identification of various materials- plastic, wood, metals-ferrous and nonferrous, rubber, glass etc.
2. Use of measuring tools for engineering applications
3. Fitting job: Detailed drawing of work piece, use of fitting tools and job preparation.
4. Carpentry job: Detailed drawing of work piece, use of carpentry tools and job preparation.
5. Sheet metal job: Detailed drawing of work piece, use of sheet metal working tools and job preparation.
6. Plumbing job: Internal/External threading, piping network using Tees, Elbows, Reducer, Bends etc
7. Hands on training on mini lathe and milling machine
8. Demonstration of welding, brazing and soldering
9. Soldering for PCB
10. 3D printing using polymer and metal.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 -Define workshop safety and various engineering materials.
CO2 -Understand various measuring equipment.
CO3 -Apply various workshop tools in preparing job for carpentry, fitting, sheet metal and plumbing.
CO4 -Examine various manufacturing operations like welding and machining.
CO5 -Evaluate soldering operation for PCB.
CO6 -Create porotype using 3D printing.

TEXT / REFERENCE BOOKS

1. Hajra Choudhury, S. K. (2008). *Elements of workshop technology* (Vols. 1–2). Media Promoters and Publishers.
2. Bawa, H. S. (2009). *Workshop practice*. Tata McGraw-Hill Education.
3. Kalpakjian, S., & Schmid, S. R. (2014). *Manufacturing engineering and technology* (7th ed.). Pearson Education.
4. Rao, P. N. (2013). *Manufacturing technology* (Vols. 1–2). Tata McGraw-Hill Education.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Evaluation based on the class performance and Laboratory book **50 Marks**
End semester examination and Viva-voce **50 Marks**

24ME102P					Engineering Graphics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	4	2	4	-	-	-	50	50	100

COURSE OBJECTIVES

- To cover the fundamental of engineering drawing and standards used in drawing
- To explain the students to communicate ideas using orthographic and isometric projection methods.
- To help students to use CAD software to prepare drawings.
- To demonstrate the presentation of drawing using sketching and 3D modelling in CAD tool.

LIST OF EXPERIMENTS

1. Introduction to Engineering Graphics: Principles of engineering graphics and their significance, drawing instruments & accessories, lettering and numbering, types of lines, dimensioning methods, basic geometric drawing, reading a drawing.
2. Orthographic Projection: Introduction to projection, types of projection, 1st angle and 3rd angle projection
3. Isometric Projection: Principles of isometric projection – isometric scale, isometric views, conventions, conversion of isometric views to orthographic views and vice-versa
4. Projection of Solids and Development of Surface: Classification of solids, projections of solids like cylinder, cone, pyramid, and prism with its inclination to reference plane, development of surfaces of right regular solids - prism, pyramid, cylinder and cone.
5. Introduction of Computer Aided Engineering Drawing: Demonstrating knowledge of the theory of CAD software, use of software in drawing, CAD software user interface, commands, Coordinate System, menus and toolbars, planes, dimensioning, saving of files, Select and erase objects, zoom tools, and others
6. Basic sketching using CAD tool: Sketch entities using tools – origin, points, lines, circle, arcs, polygons, fillets and chamfer, trim, extend and offset, copy, cut, delete and others
7. Advanced sketching using CAD tool: Sketching entities using relation constrains, Mirror, Patterning, full definition of drawing and others
8. Basic 3D modelling using CAD tool: Extrude, cut, drawing on different planes, editing, symmetric, revolving, and others
9. Computer aided drawing sheets: Preparing drawing sheets, creating different views, section view, drawing templates, and others

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 -Recalling the fundamentals of engineering graphics by considering basic rules of drawing, dimensioning, and labelling.
CO2 -Explain the principle of projection using orthographic and isometric projection.
CO3 -Represent the 2-dimensional drawing using CAD tool.
CO4 -Construct the 3-dimensional geometries using CAD tool.
CO5 -Apply the concept of engineering drawing by organizing drawing views and applying necessary dimensions by preparing drawing sheets.
CO6 -Analyse the intricate details of solid using projection of solid, sectioning of solid and development of lateral surfaces.

TEXT / REFERENCE BOOKS

1. Bhatt, N. D., Panchal, V. M., & Ingle, P. R. (2014). *Engineering drawing* (53rd ed.). Charotar Publishing House.
2. Shah, P. J. (2015). *Engineering graphics.* S. Chand Publishing.
3. Agrawal, B., & Agrawal, C. M. (2015). *Engineering drawing*. Tata McGraw-Hill Education.
4. Hanifan, R. (2014). *Perfecting engineering and technical drawing*. Springer International Publishing.
5. Autodesk. (2024). *AutoCAD user guide and reference manual*. Autodesk.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Evaluation based on the class performance and Laboratory book **50 Marks**
End semester examination and Viva-voce **50 Marks**

24PE101T					Geology for Petroleum Engineers					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- Demonstrate the fundamentals of the origin of the earth.
- Enhance skill to perform study of rocks.
- Improve skills in interpreting geological activities.
- Develop skills to identify the petroleum potential of an area.

Unit I: Physical Geology **Hours: 09**

Origin of the solar system with earth formation - stellar nucleosynthesis. The Earth's structure - crust, mantle and core; Plate tectonics and basin formation. Geological time scale; radioactive dating. Sea level change.

Unit II: Weathering and Structural Geology **Hours: 10**

Weathering - Goldich stability series; Physical or mechanical weathering; Chemical weathering; Biological weathering. Stress and strain in rocks; Strike and Dip; Fold nomenclature; Fault nomenclature; Types of joints in geology

Unit III: Mineralogy And Petrology **Hours: 10**

Crystallography -Crystal structure and unit cell types; chemical mineralogy, Bio mineralogy, Optical mineralogy; silicate mineralogy, Igneous Petrology-rock cycle, Classification of rocks; Bowen's reaction series, Formation of igneous rocks, Classification of igneous rocks, Texture of igneous rocks, Sedimentary rocks formation; structure and texture of sedimentary rocks; classification of sandstone, Folk's textural classification of carbonate sediments; Dunham's limestone classification. Formation of metamorphic rocks, Foliated rocks, and Non-foliated rocks.

Unit IV: Paleontology And Stratigraphy **Hours: 10**

Palaeontology-body fossils and trace fossils; Types of fossils; benthic fossils; pelagic fossils; Uses of fossils; Index fossils, importance of fossils in petroleum geology, Stratigraphy-Types of Stratigraphy; Laws of Stratigraphy; Unconformities; importance of stratigraphy in oil and gas traps.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand and correlate the different earth processes.
CO2 - Differentiate and classify the other minerals and rocks
CO3 - Estimate the bearing of various physicochemical and mineralogical parameters on the geo-mechanical properties of the rocks.
CO4 - Evaluate the petroleum potential of an area based on the different geological structures.
CO5 - Analyze the different strata based on their fossil assemblage and evaluate the petroleum potential based on the physiography and stratigraphy.
CO6 - Understand and correlate the different earth processes.

TEXT / REFERENCE BOOKS

1. Sengupta, S. M. (1996). *Introduction to sedimentology*. Oxford & IBH Publishing.
2. Mukherjee, P. K. (2012). *A textbook of geology*. World Scientific Publishing.
3. Mahapatra, G. B. (2015). *A textbook of geology*. CBS Publishers & Distributors.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE101P					Geology for Petroleum Engineers Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- Demonstrate the unique characteristics of various rocks.
- Develop analytical skills to identify various rock specimens.
- Imbibe skills to prepare geological maps.
- Enhance the ability to perform geological studies before reservoir modelling

LIST OF EXPERIMENTS

1. Study of crystal: clinographic projection of cube
2. Study of common silicates in hand specimens: quartz, feldspar, mica, garnet, kyanite, sillimanite
3. Study of other minerals in hand specimen: graphite, gypsum, barite, calcite, aragonite, corundum, haematite
4. Study of common igneous rocks in hand specimens: granite, basalt, rhyolite, syenite, dolerite, gabbro
5. Study of common metamorphic rock in hand specimen: slate, phyllite, schist (mica, talc, chlorite), gneiss, marble, quartzite
6. Study of common sedimentary rock in hand specimen: shale, siltstone, sandstone, conglomerate, breccia, limestone
7. Study of thin sections of minerals: quartz, feldspar, mica, pyroxene & amphiboles
8. Study of thin sections of igneous rocks: granite, gabbro, diabase, diorite, pegmatite, and peridotite.
   Study of thin sections of volcanic rocks: basalt, obsidian, pumice, tuff, rhyolite, scoria, dacite, and andesite
9. Study of thin sections of metamorphic rocks: schist, gneiss, quartzite, Novaculite or Arkansas Stone & Lapis Lazuli
10. Study of thin sections of sedimentary rocks: shale, sandstone & limestone
11. Study of selected fossils in hand specimen
12. Study of topographic maps
13. Study of geological maps

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 -Identify and classify different rock-forming minerals.
CO2 -Identify and classify various igneous, sedimentary and metamorphic rocks.
CO3 -Interpret and differentiate between different fossil groups for reconstructing the paleo environment of the studied rocks.
CO4 -Analyse the structural maps.
CO5 -Evaluate the structural deformation in the map area.
CO6 -Correlate the rock types and geological structures with some aspects of petroleum systems.

TEXT / REFERENCE BOOKS

1. McGoldrick, S. (2021). *A practical guide to introductory geology*.
2. Dixon, D. (1992). *The practical geologist: The introductory guide to the basics of geology and to collecting and identifying rocks*. Lorenz Books.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Evaluation based on the class performance and Laboratory book **50 Marks**
End semester examination and Viva-voce **50 Marks**

24HS101T					English Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- Understand of the fundamental elements of communication in English language.
- Correct expression in the English language at a basic level.
- Using appropriate vocabulary, grammar, effective paragraph construction, writing in day-to-day scenarios, including digital platforms.
- To learn and apply communication skills in different public and interpersonal contexts.

Unit I: Introduction To Communication **Hours: 07**

The Fundamentals of Language and Communication, Why English? Types of Communication, Barriers to effective Communication, Digital Communication

Unit II: English Grammar and Vocabulary **Hours: 07**

Tenses, Articles, prepositions, active voice passive voice and concord, Tools for vocabulary building, Homophones, homonyms, one word substitution, antonyms, synonyms, Root Words, Prefixes and Suffixes, Connotations. Collocations, Idioms Phrases

Unit III: Receptive Skills **Hours: 06**

Listening Skills: Difference between listening and hearing, Active listening and passive listening, Types of listening, Traits of good listener, Reading Skills: Why reading is important, Effective reading techniques, Speed Reading, The SQ3R Method

Unit IV: Compendium Skills **Hours: 06**

Note Taking and Note Making: physical, digital, collective, Summarizing, Creating e-content

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1-Demonstrate and understanding of the fundamentals of communication, including the basic components of the communication process and its significance in personal, academic, and professional contexts.

CO2-Apply grammatical rules accurately in written and spoken communication to enhance clarity, coherence, and precision.

CO3-Employ effective strategies for vocabulary development, such as word association, context clues, and mnemonic devices.

CO4-Enhance their ability to identify and interpret implicit meanings, inference, and figurative language in both written and spoken texts, contributing to deeper comprehension and critical analysis.

CO5-Develop strong written communication skills to effectively convey complex ideas and information in a clear, concise, and coherent manner within the compendium.

CO6-Apply ethical communication principles and standards of academic integrity when creating and sharing compendiums and avoiding plagiarism.

TEXT / REFERENCE BOOKS

1. Kumar, K. B. (2022). *Effective communication skills*. Khanna Book Publishing.
2. Wood, F. T. (2007). *Remedial English grammar*. Macmillan.
3. Sharma, S., & Mishra, B. (2009). *Communication skills for engineers and scientists*. PHI Learning.
4. Kaul, A. (2006). *Business communication*. Prentice-Hall of India.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24HS102T					Universal Human Values					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the need of nurturing human values through the process in value-based education system
- To understand and develop a holistic perspective on self-exploration and being in harmony with family, society and nature
- To facilitate the students in understanding harmony at all the levels and applying in their profession and work place to lead an ethical life.

Unit I: Human Values And Process of Value Education **Hours: 04**

Human values, human aspirations and the ultimate goal, understanding happiness and prosperity, appraise the meaning of satisfaction and happiness in current scenario, harmony and compatibility, values imbibed education system and process

Unit II: Knowing Self - Harmony With Self **Hours: 03**

Understanding self, capabilities and challenges, understanding material (physical facilities) and spiritual needs - need of mind and body, understanding body as an instrument, harmony between mind and body, synchronizing physical health and mental health, practicing healthy habits for healthier me

Unit III: Harmony In Relationship – Family, Society And Nature **Hours: 03**

Harmony in relationships, values for harmony in any human-human interaction, harmony in family, and society, trust and respect for others, self-esteem and ego, equality, equity, inclusion and liberation, concept of '*Vasudhaiva Kutumbakam',* understanding co-existence and sync with nature

Unit IV: Harmony In Profession And Ethical Behavior **Hours: 03**

Ethical human conduct, acceptance and respect, appraising the qualities of others, professional competence for enabling harmony in system and enabling universal human order, scope of eco-friendly systems, strategies to reach the harmonious ecosystem to reach Universal Human Order '*Sarvabhauma Vyavastha'*

Max <13Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1-Understand the significance of human values, its need, and process of value education.

CO2-Appraise the meaning of happiness and prosperity as short- and long-term goal of life. Understand them and in context of the current scenario

CO3-Distinguish between the mind and body, physical and spiritual wellbeing for harmony within self.

CO4- Assess the value of harmonious relationship based on trust, respect and enduring its role in all human-human relationships to build harmonious society

CO5- Understand the importance of harmony with nature and appreciate co-existence for harmonious ecosystem.

CO6- Create the perfect professional place and work environment following the ethical practices and strategize to uphold the human values at all the levels and interactions.

TEXT / REFERENCE BOOKS

1. Gaur, R. R., Sangal, R., & Bagaria, G. P. (2010). *A foundation course in human values and professional ethics*. Excel Books.
2. Nagaraj, A. (2010). *Jeevan vidya: Ek parichay*. Divya Path Sansthan.
3. Tripathi, A. N. (2009). *Human values*. New Age International Publishers.
4. Gandhi, M. K. (2018). *The story of my experiments with truth*. Fingerprint Publishing.
5. Illich, I. (1974). *Energy and equity*. The Trinity Press.
6. Schumacher, E. F. (1973). *Small is beautiful: A study of economics as if people mattered*. Blond & Briggs.
7. George, S. (1977). *How the other half dies: The real reasons for world hunger*. Penguin Books.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**